

Estimation of multiple treatment effect on software revenue

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1 Introduction

Firms routinely invest substantial resources in customer acquisition and retention strategies to increase revenue. Evaluating the effectiveness of these interventions is fundamentally a causal inference problem, as the goal is to determine whether observed changes in revenue are caused by the interventions rather than driven by confounding factors or selection effects.

In this applied project, we demonstrate how causal inference methods can be used to estimate treatment effects and construct optimal treatment allocation policies. The analysis is based on data from a software company that offers two types of interventions to its customers: a discount and technical support. The objective is to estimate the causal impact of these treatments on revenue and leverage these estimates to design an optimal treatment assignment to maximize expected returns.

2 Data and Methodology

The data used in this project comes from a publicly available Kaggle dataset (Software Usage Promotion Campaign Uplift Model). Each observation corresponds to a customer (i.e., another company), with the outcome defined as the revenue generated from that customer. The dataset includes two binary treatments (discount and technical support) and eight covariates describing firm characteristics, such as company size and IT expenditure. A detailed description of all variables is provided in Table 1. The dataset contains 2,000 observations. Since the treatments are binary, we consider the four possible treatment combinations (no treatment, discount only, technical support only, and both treatments) as distinct treatment arms in the analysis. Each of the four treatment combinations contains a similar number of observations, as shown in Table 2.

The treatment assignment mechanism is unknown; however, it is unlikely to be fully randomized. In particular, some covariates (most notably company size and IT spend) appear to be imbalanced across treatment groups, as shown in Figure 1. Furthermore, Figures 2 and 3 indicate that firms receiving both treatments tend to have higher IT expenditure and larger firm size. These patterns suggest that treatment assignment is at least partially driven by observable covariates, with larger and higher-spending firms more likely to receive both interventions. As a result, this problem is best understood as an observational study.

Figure 1 indicates that propensity score weighting methods (IPW, overlap, and treated weights) achieve good covariate balance across treatment groups. In addition, Figure 4 shows substantial overlap in the estimated propensity score distributions across groups, supporting the validity of

causal comparisons. However, some observations exhibit extremely low propensity scores, particularly in the “No treatment” propensity scores. This can lead to highly variable inverse probability weights and unstable estimates. To address this issue, we apply optimal trimming as implemented in the `PSweight` package [2]. Figure 5 shows that covariate balance remains satisfactory after trimming. Figure 6 indicates that the propensity score distributions are similar to the untrimmed case, but with extreme observations removed. The counts of trimmed observations by group are shown in Table 3. This improves the stability of IPW by reducing the influence of units with very low propensity scores. We estimate causal effects using IPW, overlap, ATT, and doubly robust estimators on the full sample, and additionally apply IPW and doubly robust methods on the trimmed sample. Standard errors are obtained via bootstrap with $R = 100$ replications. Since there are four treatment arms, there are six possible pairwise contrasts. We use the weighting methods described above to estimate the causal effects for each of these pairwise comparisons.

It is also of interest to understand how customer characteristics influence treatment effects. To this end, we estimate the Conditional Average Treatment Effect (CATE), which captures how the effect of each treatment varies with covariates. This allows us to identify customers who are expected to generate higher revenue under specific interventions and to design targeted strategies to maximize returns.

To estimate heterogeneous treatment effects, we use a multi-arm causal forest implemented in the `grf` package [1]. In this approach, the outcome regression $m(X) = \mathbb{E}[Y \mid X]$ and the generalized propensity score $e(X) = \mathbb{E}[W \mid X]$ are first estimated using separate random forests. Treatment effects are then obtained by regressing residualized outcomes on residualized treatments.

Using “No treatment” as the baseline arm, the model provides estimates of the CATE for each of the remaining treatment groups relative to this baseline. In particular, for each company we estimate the effects of “Both treatments,” “Discount only,” and “Tech support only” compared to “No treatment.”

This causal forest model allows us to assess feature importance in explaining heterogeneity in treatment effects, highlighting which covariates are most relevant for variations in the CATE and informing the design of revenue-maximizing strategies. A strategy to use the CATE predictions to optimize treatment assignment to maximize revenue is described in section 4.

3 Results

The results from the propensity score weighting methods are reported in Tables 4 to 9. Across all contrasts, the different estimators produce broadly consistent estimates, which increases confidence in the robustness of the results. A notable pattern is that the doubly robust (DR) estimator consistently yields the lowest standard errors, while trimming generally reduces variability and improves stability of the estimates.

For instance, in Contrast 2 (Table 5), which estimates the average treatment effect

$$\mathbb{E}[Y(\text{Both treatments}) - Y(\text{No treatment})],$$

the trimmed DR estimator yields an estimate of approximately \$11,810. In contrast, a naive comparison of observed mean revenues between these two groups gives a difference of approximately \$20,198. This substantial discrepancy highlights the importance of adjusting for confounding, as the unadjusted difference substantially overstates the causal effect.

Regarding the causal forest results, Figure 7 shows that the estimated CATEs vary substantially across observations and treatment contrasts, indicating strong heterogeneity in treatment effects. This suggests that firm characteristics play an important role in determining the revenue impact of each intervention, even under the same treatment assignment.

Figure 8 reports the causal forest feature importance measures. It is evident that company size and IT spend are the most important covariates driving heterogeneity in treatment effects. Notably, these are also the covariates that exhibit the largest imbalance across treatment groups, further reinforcing the limitations of direct comparisons between treated and untreated units.

Figure 9 shows that the effect of the treatments relative to no treatment increases with IT spend, while Figure 10 shows a similar pattern with respect to firm size. This implies that larger firms and firms with higher IT spending tend to benefit more from the interventions.

As shown earlier in Figures 2 and 3, these same types of firms are more likely to receive both treatments, suggesting that the company may already be targeting high-value customers. However, the causal forest model allows us to move beyond this targeting strategy by providing individualized CATE estimates for each treatment arm. This enables a more refined treatment assignment mechanism that optimally allocates treatments under a budget constraint.

4 Treatment assignment optimization

There are several ways to exploit the causal forest model for decision-making. We consider a setting in which treatments have associated costs: offering a discount costs \$6000, technical support costs \$7000, so offering both treatments costs \$13000. The no-treatment option has zero cost. The objective is to determine an optimal treatment allocation rule that maps covariates to treatments in order to maximize expected net revenue subject to a total budget constraint.

Under the observed assignment, the total cost of treatments in the data is \$13,168,000.

Using the estimated CATEs, we assign treatments based on covariates to maximize expected net value. We define the unit-level net value of treatment k as

$$\text{net}_k(X) = \tau_k(X) - c_k,$$

where c_k denotes the cost of treatment k , and $\tau_k(X) = \mathbb{E}[Y(k) - Y(0) \mid X]$ is the conditional average treatment effect relative to the no-treatment baseline. Thus, $\text{net}_k(X)$ represents the expected incremental profit of assigning treatment k to a unit with characteristics X .

The optimal assignment is then obtained by solving the constrained optimization problem:

$$\arg \max_{d_{ik}} \sum_{i=1}^n \sum_k d_{ik} \text{net}_k(X_i), \quad \text{s.t.} \quad \sum_k d_{ik} = 1, \quad \sum_{i,k} d_{ik} c_k \leq B,$$

with d_{ik} in $\{0, 1\}$, which is a binary decision variable equal to 1 if unit i is assigned to treatment k , and 0 otherwise. B is the maximum total cost allowed.

Finally, we evaluate the optimized treatment assignment by comparing its expected performance against the observed treatment assignment using the same CATE-based net value framework. The expected gains from the optimized allocation is \$4,307,291, while the expected gains under the observed allocation is \$1,225,866. This shows that using the causal forest model to learn a optimize the treatment allocation has a big potential of increasing the revenue. For future observations, we could use the causal forest to estimate the CATEs based on their covariates, and then solve the optimization problem to assign treatments in a way that maximizes expected net profit under the budget constraint.

Reproducibility

All code and analysis are available at: [Reproducible Code \(R Markdown\)](#)

Video Presentation

Video presentation: [Watch Video](#)

A Tables

Name	Type	Description
Global Flag	Covariate	Whether the customer has global offices
Major Flag	Covariate	Whether the customer is a large enterprise
SMC Flag	Covariate	Whether the customer is a Small Medium Corporation
Commercial Flag	Covariate	Whether the customer's business is commercial (vs public sector)
IT Spend	Covariate	Amount spent on IT-related purchases (\$)
Employee Count	Covariate	Number of employees
PC Count	Covariate	Number of PCs used by the customer
Size	Covariate	Customer size measured by annual revenue
Tech Support	Treatment	Whether the customer received technical support (binary)
Discount	Treatment	Whether the customer received a discount (binary)
Revenue	Outcome	Revenue generated from the customer (\$ software purchases)

Table 1: Description of variables in the dataset

	Discount = 0	Discount = 1
Tech Support = 0	517	477
Tech Support = 1	462	544

Table 2: Joint distribution of treatment assignments

	Both treatments	Discount only	No treatment	Tech Support only
Original sample	544	477	517	462
Trimmed sample	73	12	3	11
Remaining	471	465	514	451

Table 3: Number of observations by treatment group before and after trimming

Method	Trimmed	Estimate	Std. Error	95% CI
IPW	Untrimmed	7199.76	316.66	[6464.38, 7750.44]
IPW	Trimmed	6908.40	209.06	[6462.82, 7269.08]
Overlap	Untrimmed	7073.33	238.33	[6627.48, 7506.83]
ATT	Untrimmed	7166.72	298.67	[6504.19, 7682.54]
DR	Untrimmed	7142.89	84.51	[6946.37, 7291.77]
DR	Trimmed	6847.51	84.08	[6687.35, 7006.08]

Table 4: Contrast 1: $\mathbb{E}[Y(\text{Both treatments}) - Y(\text{Discount only})]$

Method	Trimmed	Estimate	Std. Error	95% CI
IPW	Untrimmed	13405.60	403.30	[12578.60, 14022.64]
IPW	Trimmed	12222.98	219.14	[11838.74, 12628.81]
Overlap	Untrimmed	12066.75	255.03	[11506.76, 12492.92]
ATT	Untrimmed	12785.37	465.34	[11950.62, 13590.02]
DR	Untrimmed	12917.68	189.91	[12542.92, 13242.05]
DR	Trimmed	11810.44	134.87	[11534.24, 12068.86]

Table 5: Contrast 2: $\mathbb{E}[Y(\text{Both treatments}) - Y(\text{No treatment})]$

Method	Trimmed	Estimate	Std. Error	95% CI
IPW	Untrimmed	5664.79	324.96	[5165.58, 6302.42]
IPW	Trimmed	5041.97	191.80	[4692.02, 5409.00]
Overlap	Untrimmed	5050.75	256.38	[4538.97, 5485.11]
ATT	Untrimmed	5568.17	323.57	[4849.12, 6064.08]
DR	Untrimmed	5615.33	126.34	[5368.35, 5834.28]
DR	Trimmed	4870.73	115.57	[4662.09, 5090.98]

Table 6: Contrast 3: $\mathbb{E}[Y(\text{Both treatments}) - Y(\text{Tech support only})]$

Method	Trimmed	Estimate	Std. Error	95% CI
IPW	Untrimmed	6205.83	491.39	[5257.82, 7114.69]
IPW	Trimmed	5314.58	226.46	[4917.59, 5755.88]
Overlap	Untrimmed	4993.42	183.88	[4599.99, 5343.05]
ATT	Untrimmed	5618.65	454.02	[4953.82, 6624.21]
DR	Untrimmed	5774.80	145.61	[5490.81, 6058.05]
DR	Trimmed	4962.93	96.96	[4776.01, 5143.65]

Table 7: Contrast 4: $\mathbb{E}[Y(\text{Discount only}) - Y(\text{No treatment})]$

Method	Trimmed	Estimate	Std. Error	95% CI
IPW	Untrimmed	-1534.97	472.81	[-2268.36, -476.41]
IPW	Trimmed	-1866.43	188.09	[-2200.67, -1461.57]
Overlap	Untrimmed	-2022.58	144.02	[-2300.38, -1734.01]
ATT	Untrimmed	-1598.54	278.60	[-2057.72, -934.92]
DR	Untrimmed	-1527.56	84.48	[-1709.61, -1379.37]
DR	Trimmed	-1976.77	87.14	[-2142.33, -1810.67]

Table 8: Contrast 5: $\mathbb{E}[Y(\text{Discount only}) - Y(\text{Tech support only})]$

Method	Trimmed	Estimate	Std. Error	95% CI
IPW	Untrimmed	7740.80	496.43	[6641.52, 8547.52]
IPW	Trimmed	7181.01	131.41	[6874.88, 7401.97]
Overlap	Untrimmed	7016.00	117.79	[6771.45, 7215.77]
ATT	Untrimmed	7217.20	276.92	[6672.65, 7726.81]
DR	Untrimmed	7302.36	110.15	[7052.16, 7494.73]
DR	Trimmed	6939.71	74.88	[6808.29, 7102.27]

Table 9: Contrast 6: $\mathbb{E}[Y(\text{Tech support only}) - Y(\text{No treatment})]$

B Figures

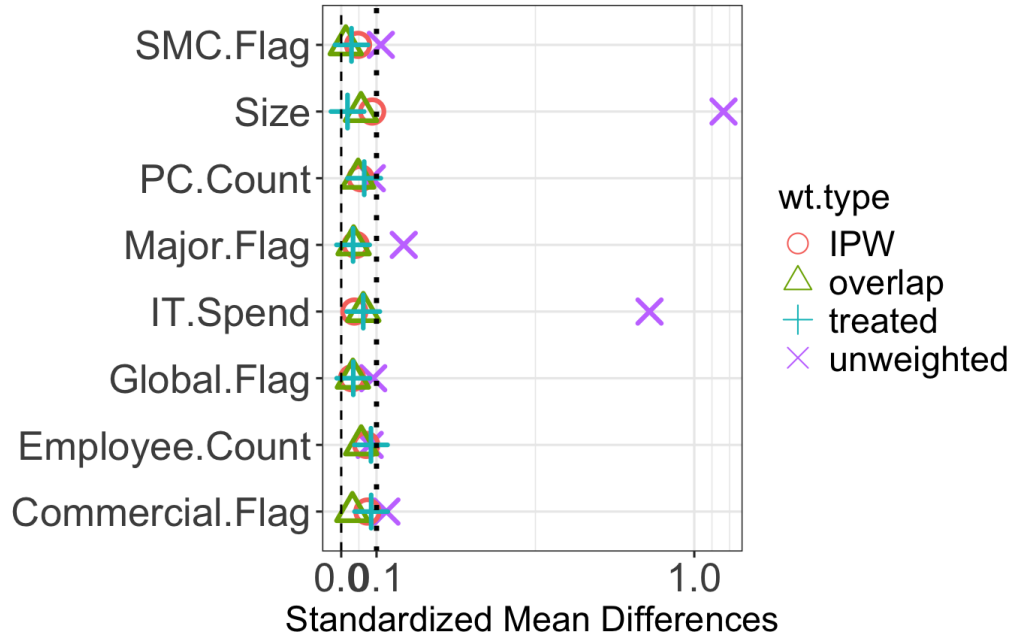


Figure 1: Covariate balance across treatment groups.

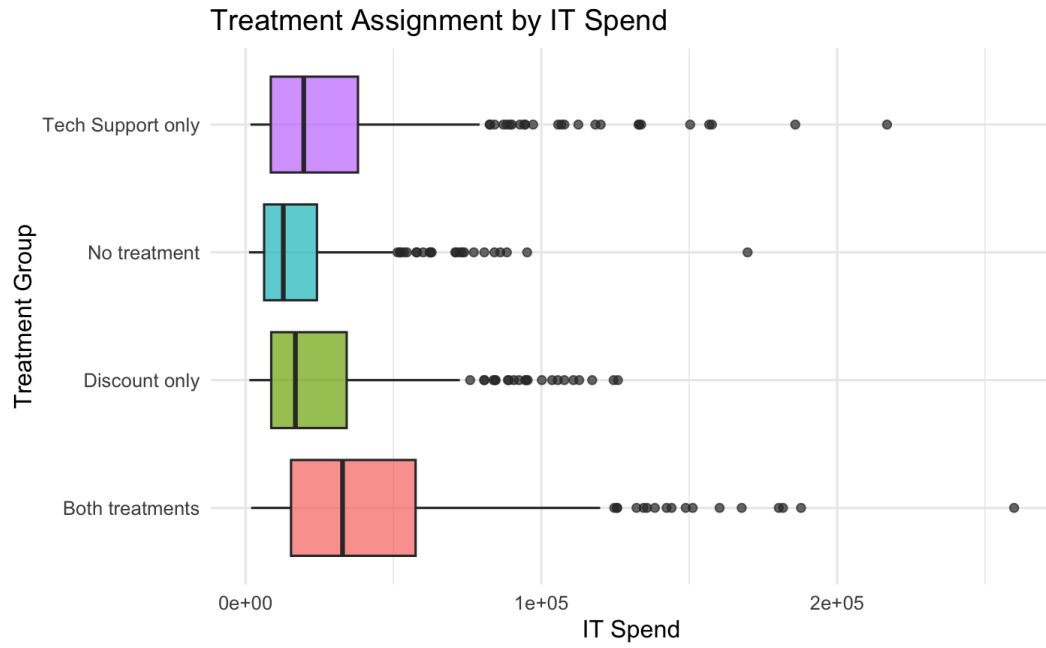


Figure 2: Treatment assignment by IT spend.

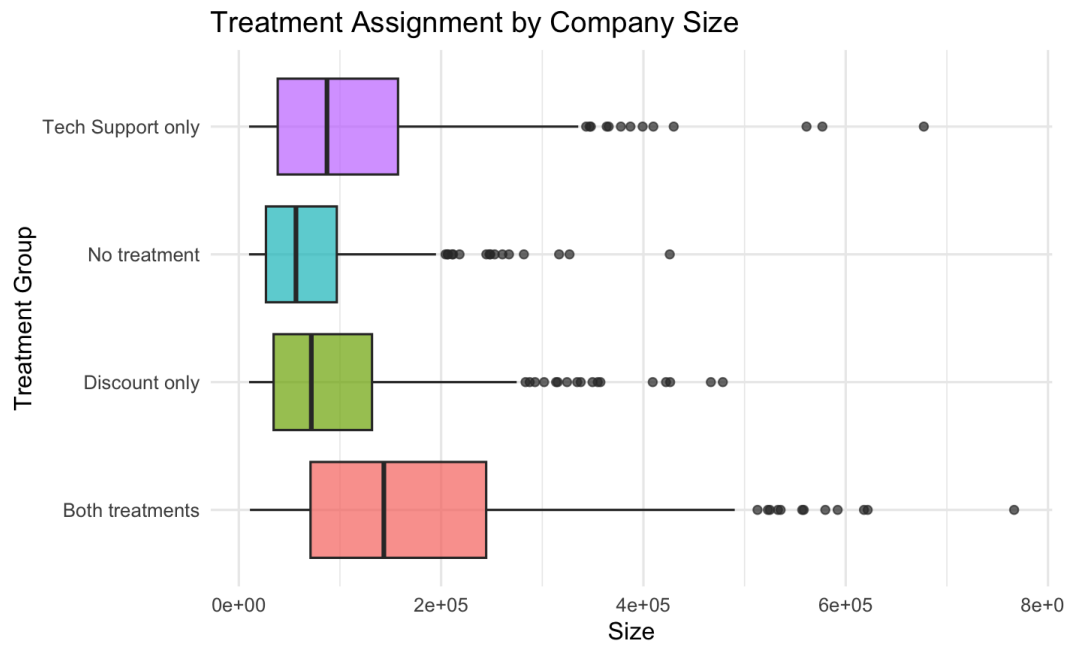


Figure 3: Treatment assignment by Size.

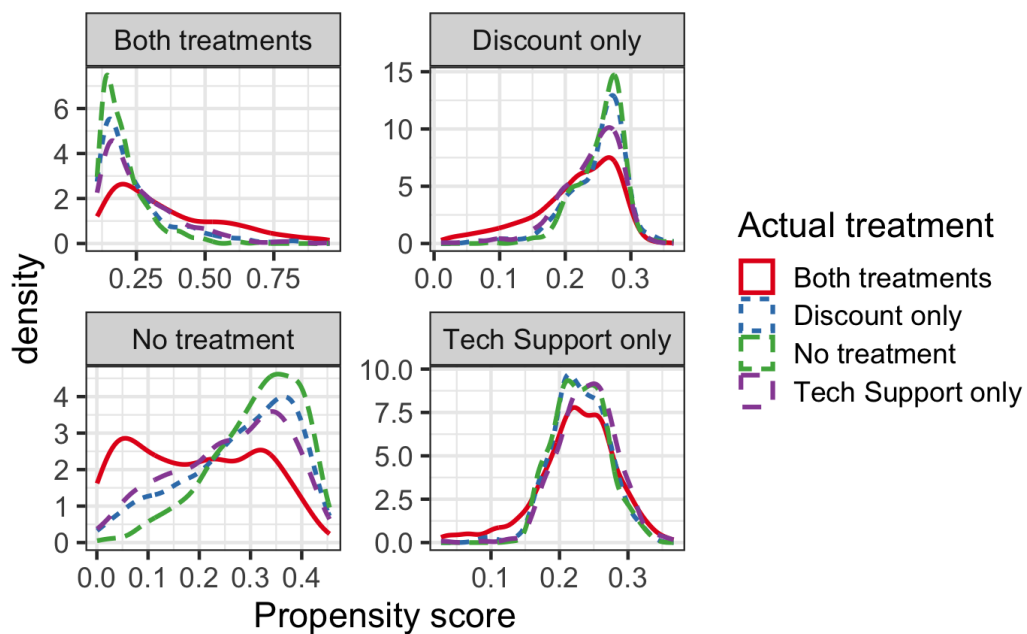


Figure 4: Propensity scores densities by treatment group.

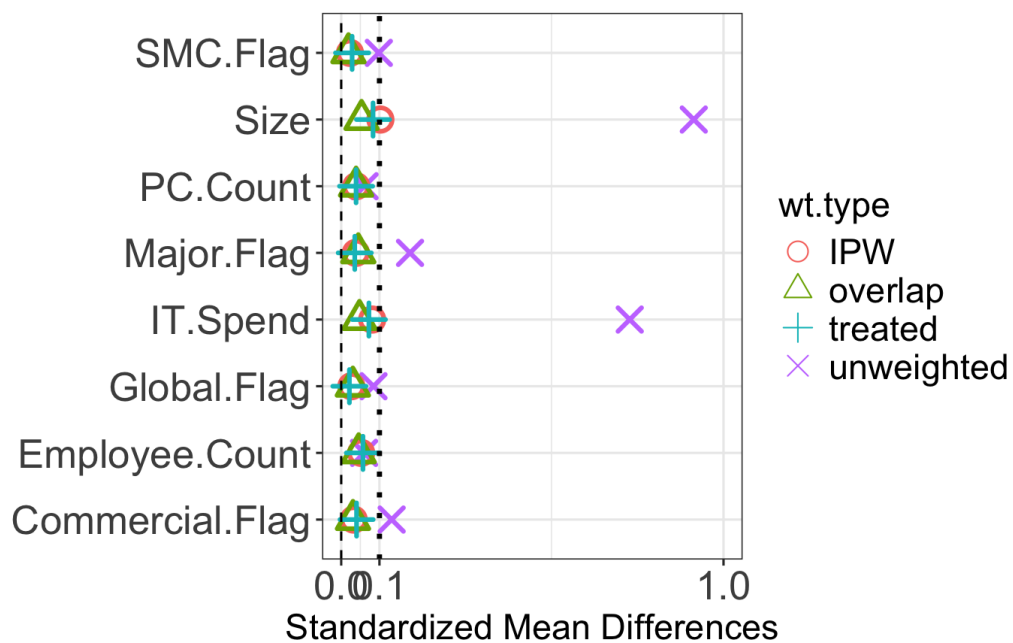


Figure 5: Covariate balance across treatment groups with trimmed data.

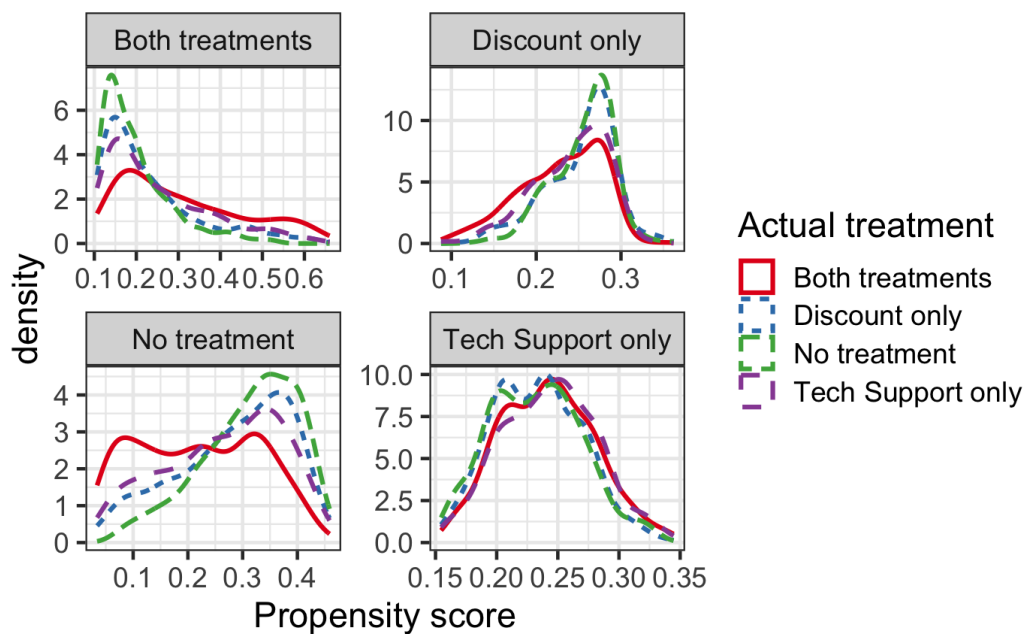


Figure 6: Propensity scores densities by treatment group with trimmed data.

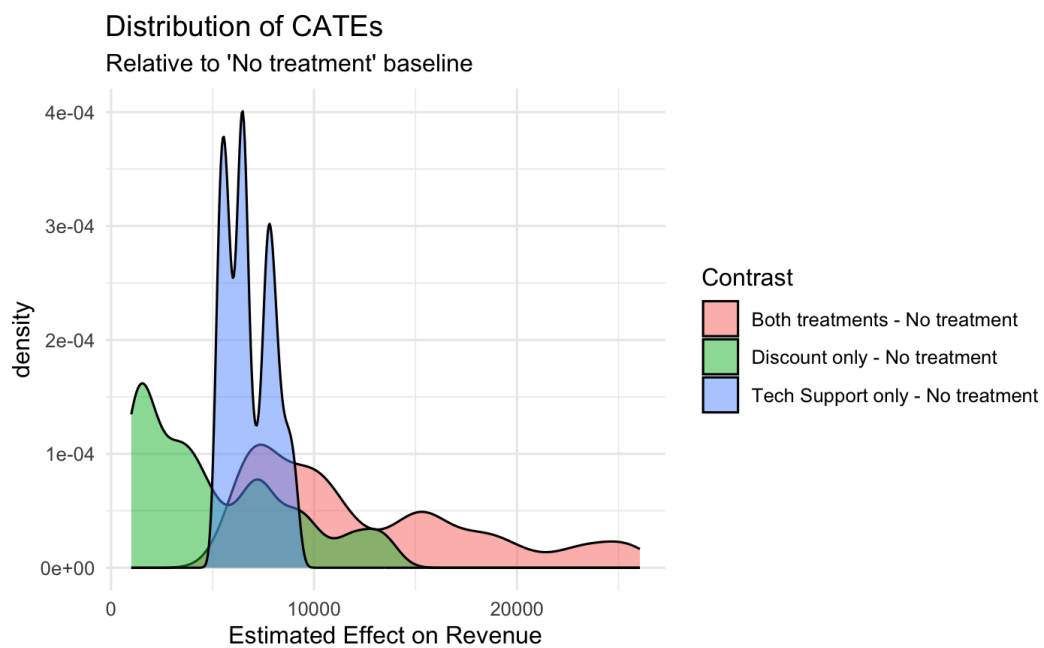


Figure 7: Distribution of CATE by contrast.

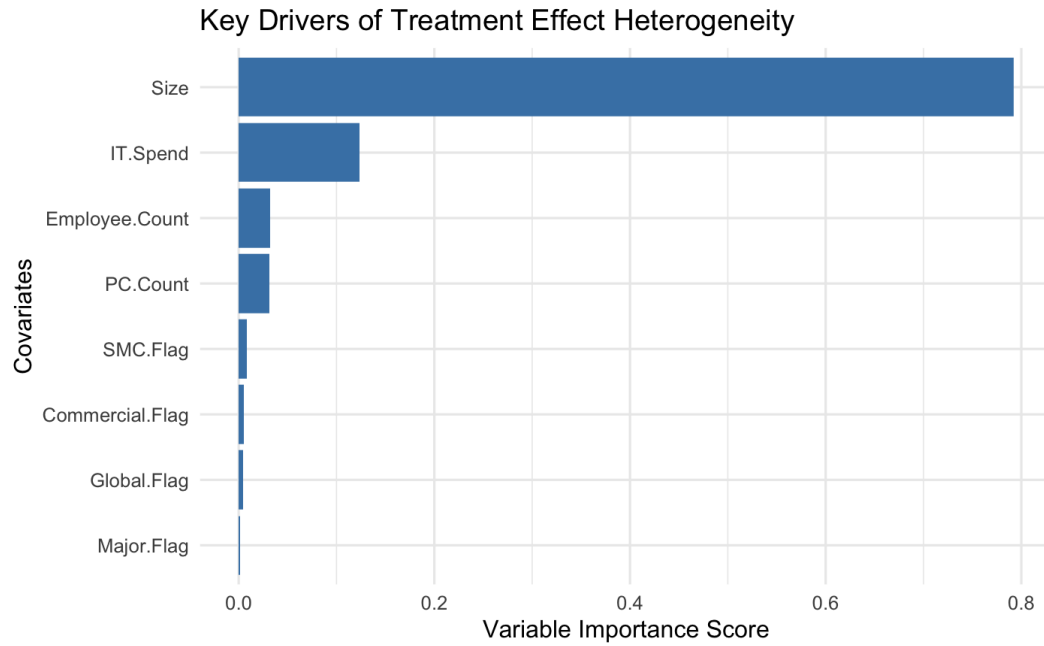


Figure 8: Causal forest feature importance.

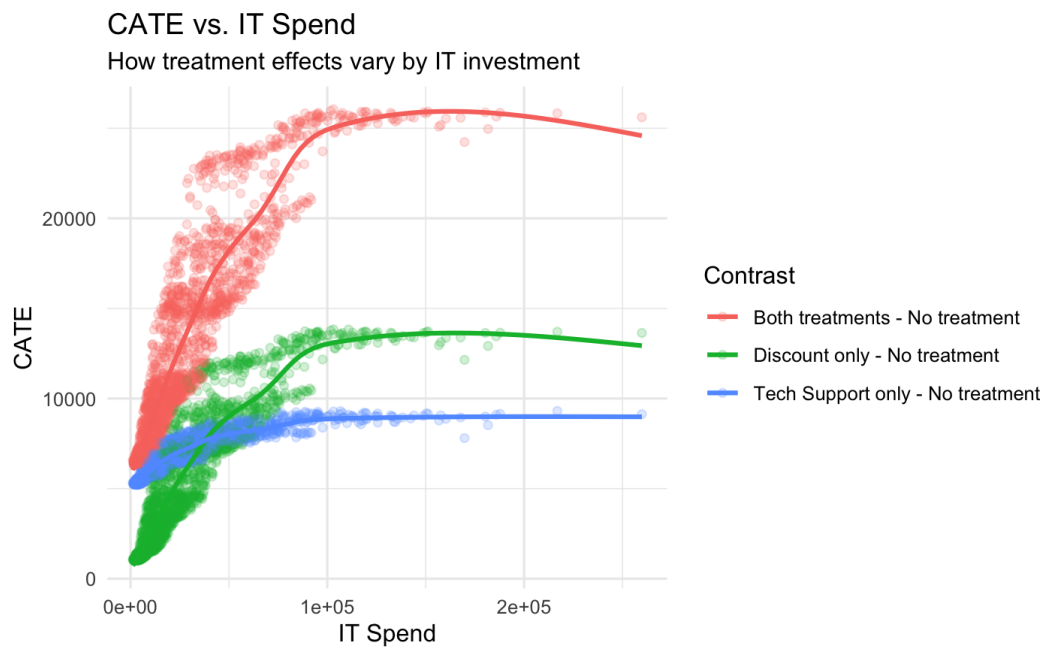


Figure 9: IT Spend vs estimated CATE.

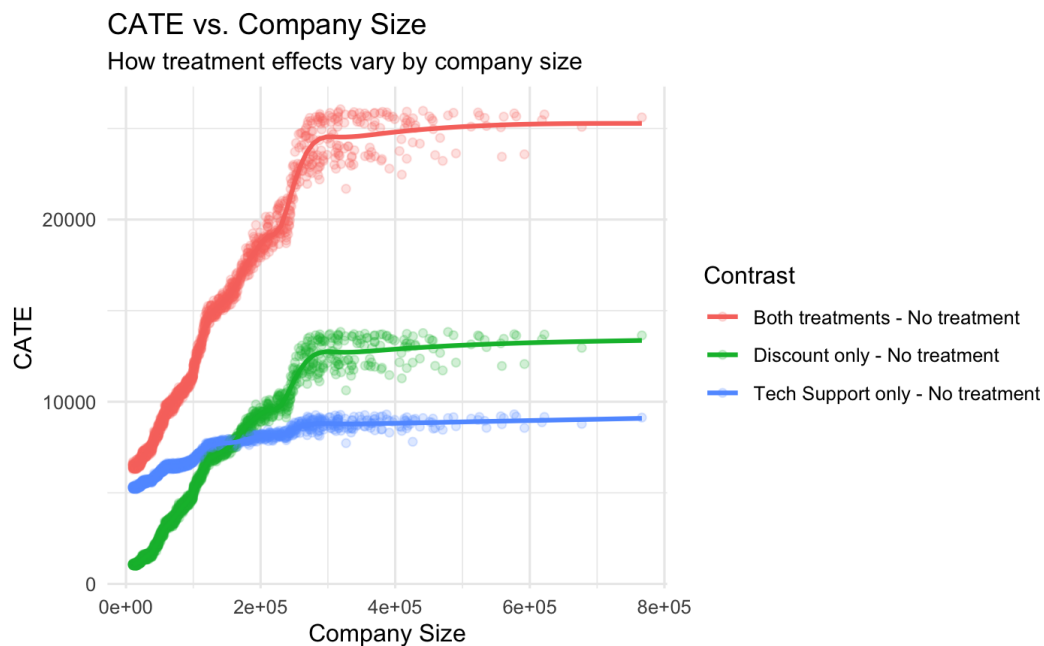


Figure 10: Size vs estimated CATE.

References

- [1] Julie Tibshirani, Susan Athey, Erik Sverdrup, and Stefan Wager. *grf: Generalized Random Forests*, 2026. R package version 2.6.1.
- [2] Tianhui Zhou, Guangyu Tong, Laine Thomas, and Fan Li. *PSweight: Propensity Score Weighting for Causal Inference with Observational Studies and Randomized Trials*, 2025. R package version 2.1.2.